

DEGAiN TRAINING

BAYESIAN POPULATION STRUCTURE IN DEC

A practical workbook: Inferring Bacterial Population Structure from Genomes

Overview

This workbook guides participants through the complete workflow of inferring bacterial population structure from annotating high-quality bacterial genomes and constructing pangenome to carrying out Bayesian population structure for the identification of biologically meaningful populations for downstream evolutionary and epidemiological analyses.

Learning objectives

On completing this practical session, participants will be able to:

- Annotate bacterial genomes using **bakta**
- Construct a pangenome and generate a core genome alignment using **panaroo**.
- Infer hierarchical bacterial population structure using **FastBAPS**.
- Relate population structure to phylogeny and epidemiological metadata
- Visualise and interpret bacterial population structure using **microreact**

Directory Structure on the server

All practical exercises for this workshop will be performed within the **degain** project directory. Each training module has its own subdirectory. For this session, we will be working in the **baps** directory.

Within the **baps** directory, participants have been assigned to one of three groups: **group1**, **group2** and **group3**

Each group directory contains a **genomes** folder with the assembled genome sequences (FASTA format) that will be used for the practical exercises.

Navigate to the **baps** working directory:

```
$ cd /data/degain/baps
```

View the available group directories:

```
$ ls
```

Change to your assigned group directory (example: **group2**):

```
$ cd groupB
```

List the contents of the **genomes** directory to confirm the genomes available for analysis:

```
$ ls genomes
```

You should see **20 assembled genome files** in FASTA format, which will be used for this practical session.

1. Genome annotation

Objective: To convert each assembled genomes to a standardized set of predicted genes and genomic features.

For this, we will use a tool called **Bakta**, a rapid bacterial genome annotation tool that produces standardized genome feature predictions suitable for comparative genomics and pangenome analysis.

<https://github.com/oschwengers/bakta>

Inputs and commands

Ensure you are in your assigned working directory before starting the analysis.

```
$ cd /data/degain/baps/group2
```

Activate the Bakta conda Environment

```
$ conda activate degain_bakta
```

Confirm that Bakta is installed and accessible.

```
$ bakta -version
```

Display the available commands and options.

```
$ bakta -help
```

Run Bakta

```
$ for genome in genomes/*.fasta; do base=$(basename "$genome"
.fasta); bakta "$genome" --db /home/oakeke/local_data/bioinfo/db -
-output bakta_output/"$base" --prefix "$base" --genus Escherichia
--species coli --threads 8; done
```

What the script does. For every genome in the dir genomes, it executes `bakta -db bakta_db --output bakta_output/genome --prefix <genome> --threads N`. Here `--db` is the annotation database, `--prefix` names outputs after the isolate for traceability, and `--threads` sets CPU cores. Results are written to `bakta_output/<genome>/.`

Expected Output

Bakta generates several output files in different formats. A complete description of each output file is available in the official Bakta documentation: <https://github.com/oschwengers/bakta>

For this practical, the most important output is the **.gff3** file, as it will serve as the input for the pangenome analysis in the next session.

The **.gff3** file contains the genome feature predictions, including the coordinates and annotations of coding sequences (CDSs), tRNAs, rRNAs, and other genomic features for downstream comparative genomic analyses. The **.txt** file is the quality-control summary. You can open any of the `<genome>` **.txt** file see exactly what Bakta reports.

2. Pangenome & Core genome alignment

Objective. Determine which genes are shared across isolates and build the core-gene alignment.

In this practical session, we will use **Panaroo** to cluster annotated genes into gene families and classify them according to their frequency across the dataset. The **core** (present in all or nearly all isolates) and accessory classified into **shell** (present in many but not all isolates), and **cloud** (rare accessory genes present in only a few isolates). A core genome alignment will be generated as well. For detailed documentation: <https://gthlab.au/panaroo/#/>

Input and commands

Required input: Bakta-generated GFF3 files.

Note: since each genome was annotated separately, first create a text file containing the path to every **.gff3** file. Each line in the text file should correspond to one isolate.

Ensure that you are in the working directory:

```
$ cd /data/degain/baps/groupB
```

Generate the GFF3 pathway file:

```
$ find bakta_output -type f -name "*.gff3" | sort >
gff3_pathway.txt or
$ ls bakta_output/*//*.gff3 > gff3_pathway.txt
```

The **find** command searches the **bakta_output** directory and its subdirectories for files ending in **.gff3**. The paths are sorted and saved in **gff3_pathway.txt**.

Check the number of GFF3 files listed:

```
$ wc -l gff3_pathway.txt
```

The output should show 20 files.

Inspect the first few entries:

```
$ head gff3_pathway.txt
```

The file should contain one GFF3 path per line, for example:

```
bakta_output/DEC001/DEC001.gff3
bakta_output/DEC002/DEC002.gff3
bakta_output/DEC003/DEC003.gff3
```

Run Panaroo

Activate the Panaroo environment:

```
$ conda activate degain_panaroo
```

Verify that Panaroo is available:

```
$ panaroo -version
```

Run Panaroo:

```
$ panaroo -i gff3_pathway.txt -o panaroo_output --clean-mode moderate  
--remove-invalid-genes -a core --core_threshold 0.95 -t 8
```

Panaroo will group the annotated genes into gene families, classify them according to their frequency across the isolates, and construct a core genome alignment.

Expected output

Panaroo writes many files into panaroo_output/, only core_gene_alignment.aln is used for Bayesian population structure. The core_gene_alignment.aln is a concatenated alignment file of the core genome.

Task Record the core, shell and cloud gene counts. State one biological reason the core might be smaller than expected.

3. Bayesian population structure

Objective: Determine population structure

In this practical session, we will use **FastBAPS** to infer genetically related populations from the core genome alignment generated by Panaroo. FastBAPS groups genomes based on patterns of nucleotide variation across the core genome and identifies population structure at nested levels from broad populations (Level 1) to sub clusters (Level 2).

Input and commands

Required input: core_gene_alignment.aln

If you have navigated away from your assigned working directory, return to it. For example:

```
cd /home/degain_USER/group1_cluster12
```

Run fastbaps

Activate the fastbaps environment:

```
$ conda activate degain_fastbaps
```

```
$ Rscript fastbaps.R
```

What the script runs. an R routine that calls `import_fasta_sparse_nt` (loads the alignment as a sparse SNP matrix), `optimise_prior` (tunes how readily populations are split), and `multi_res_baps` (produces the nested cluster levels), writing `fastbaps_output/fastbaps_clusters.csv`.

Interpretation: Level 1 gives broad populations; higher levels subdivide them. Cluster numbers are identifiers which makes more biological meaning when checked against the tree and metadata.

Discussion questions

Why can two isolates with the same MLST fall into different FastBAPS population cluster? and what does this tell us about the resolution of the two approaches?

A FastBAPS population contains several sequence types. What biological processes could produce this?